

MANUFACTURING PROCESSES

II B.TECH I Semester : MECHANICAL ENGINEERING								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6ME12	PCC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	40	60	100
COURSE OVERVIEW: This course is intended to introduce various manufacturing processes such as casting, joining, forming and their relevance in current manufacturing industry. Also focus to explore processing methods of plastics.								
COURSE OUTCOMES: The course should enable the students to 1. Explain casting, casting methods, preparation of patterns and castings. 2. Enumerate various welding & cutting techniques. 3. Explain hot working, cold working, extrusion, rolling and forging processes. 4. Select equipment for various deformation processes. 5. Demonstrate various methods used for producing plastic shapes.								
UNIT-I	CASTINGS & METHODS OF MELTING							
CASTING: Steps involved in making a casting, Advantage of casting and its applications. Patterns and Pattern making, Types of patterns, Materials used for patterns, pattern allowances and their construction, Principles of Gating, Gating ratio and design of Gating systems Solidification of casting, Concept, Solidification of pure metal and alloys, short and long freezing range alloys. Risers Types function and design, special casting processes 1) Centrifugal 2) Die 3) Investment.								
METHODS OF MELTING: Crucible melting and cupola operation, steel making processes								
UNIT-II	WELDING & CUTTING OF METALS							
WELDING: Classification of welding process types of welds and welded joints and their characteristics, design of welded joints, Gas welding, ARC welding, Forge welding, resistance welding, Thermit welding and Plasma (Air and water) welding. TIG and MIG, welding, Friction welding, Induction welding, Explosive welding, Laser welding, Soldering and Brazing. Heat affected zones in welding; welding defects, causes and remedies, destructive, non-destructive testing of welds.								
CUTTING OF METALS: Oxy Acetylene Gas cutting, water plasma, cutting of ferrous, non-ferrous metals.								
UNIT-III	HOT AND COLD WORKING, EXTRUSION							
HOT WORKING AND COLD WORKING: Strain hardening, recovery, recrystallisation and grain growth, Comparison of properties of cold and hot worked parts.								
EXTRUSION OF METALS: Basic extrusion process and its characteristics. Hot extrusion and cold extrusion, Forward extrusion and backward extrusion, Impact extrusion, Hydrostatic extrusion.								
UNIT-IV	ROLLING							

ROLLING: Fundamentals, theory of rolling, types of Rolling mills and Forces in rolling and power requirements. Stamping, forming and other cold working processes: Blanking and piercing, Bending and forming,

DRAWING: Drawing and its types, wire drawing and Tube drawing, coining, Hot and cold spinning Forces and power requirement in the above operations.

UNIT-V	FORGING & PROCESSING OF PLASTICS
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FORGING: Principles of forging, Tools and dies, Types Forging, Smith forging, Drop Forging, Roll forging, Rotary forging, forging defects.

PROCESSING OF PLASTICS: Types of Plastics, Properties, applications and their Processing methods and Equipment (blow and injection modelling).

Text Books:

1. P. N. Rao (2011), Manufacturing Technology, Vol -1, 3rd edition, Tata McGraw- Hill education (P) Ltd, New Delhi
2. S. Kalpakjain (2005), Manufacturing Engineering and Technology, 4th edition, Pearson Education, New Jersey.

Reference Books:

1. R. K. Jain (2010), Production Technology, 16th edition, Khanna publishers, New Delhi, India.
2. B.S. Raghuwanshi (2011), A course in workshop Technology, Vol - II, 3rd Edition, Dhanpat Rai & Co, New Delhi, India
3. Gosh (2004), Manufacturing science, Affiliated East-west press (p) Ltd, New Delhi, India